

Module 4

Continuous Quality Improvement

Hands-on Training on Outcome-Based Education (OBE)

Board of Accreditation for Engineering and Technical Education (BAETE)
Training on OBE and Accreditation, 26-27 June 2026



Presented by

Mafin Muntasir Rahman
Assistant Professor
Dept. of EEE, BAUST

Courtesy of

Nehreen Majed, PhD
Member, Training Task Force, BAETE
Professor, Department of Civil &
Environmental Engineering, NSU



Outline of the Session

- Outcome of the session
- CQI - An Integral Component of OBE
- Components of CQI
- Inputs and Feedback from Stakeholders
- Assessment of Feedback from Stakeholders
- Analysis and Improvement Actions in CQI
- Records of Evidence
- Establish CQI Cycle
- Example and Exercise

Outcome of the Session

- Familiarity with CQI process
- Informed knowledge on CQI components
- Better preparation for CQI implementation
- Better understanding on the action items and planning requirements to close CQI loops
- Identification of the key stakeholders involved in the CQI process
- Identification of the supporting resources required to make CQI process effective to contribute to OBE implementation

CQI - An Integral Component of OBE

Continuous Quality Improvement (CQI) is an ongoing process where achievements are evaluated based on data and stakeholders' feedback to initiate an action plan for improvement.

Stakeholders

Students

Alumni

Faculty members

Employers

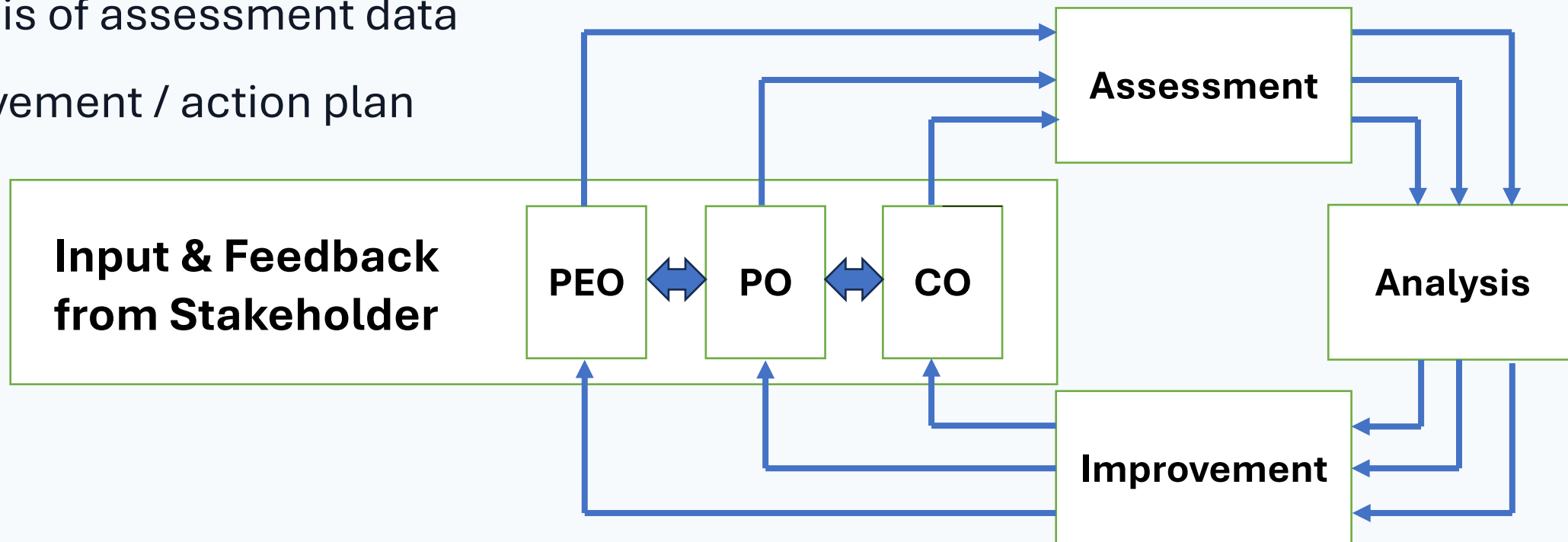
**Industry Advisory Panel
(IAP)**

Professional organizations

Components of CQI

Three components of CQI:

- Feedback from stakeholders to collect assessment data
- Analysis of assessment data
- Improvement / action plan



Inputs and Feedback from Stakeholders

Program must complete the CQI loops periodically for:

COs



- Feedback from students
- Feedback from course instructors

POs



- Feedback from graduates
- Feedback from course instructors

PEOs



- Feedback from graduates
- Feedback from employers, IAP

Curriculum and
Teaching Quality



- Feedback from students, graduates
- Feedback from faculty members, IAP

CQI loops should be a regular practice - not done occasionally just before the accreditation visit.

Student Feedback

- In every course, students must give feedback on the course including COs and constructive alignment
- Meeting with Student Advisory Board / Student Council
- Exit interviews / surveys with graduating students
- Student evaluation survey and individual faculty-student interaction

Faculty Feedback

- Feedback on course outcome using course embedded assessment, constructive alignment etc.
- Departmental/Curriculum committee meetings
- Regular faculty meetings and interactions

IAP/IAB Feedback

- Industry Advisory Panel/Board (IAP/IAB) meeting
- Surveys with industry
- Employer surveys

Alumni Feedback

- Alumni Advisory Board
- Surveys with alumni

Analysis and Improvement Actions in CQI

Analysis: Data need to be analyzed carefully to

- Evaluate achievements of PEO, PO and CO
- Identify strengths and weaknesses
- Identify opportunities and threats
- Gather the big picture, trends and patterns

Improvement: Prepare an action plan based on data analysis

- Update objectives, outcomes, contents, delivery and assessment
- Update policies
- Update infrastructural and other facilities

Curriculum Design Evidence

- Articulation of COs in core program courses
- Mapping of COs into POs
- Mapping of POs into PEOs
- Alignment of PEOs with mission statement
- Involvement of industry and alumni in curriculum design

Course File Evidence

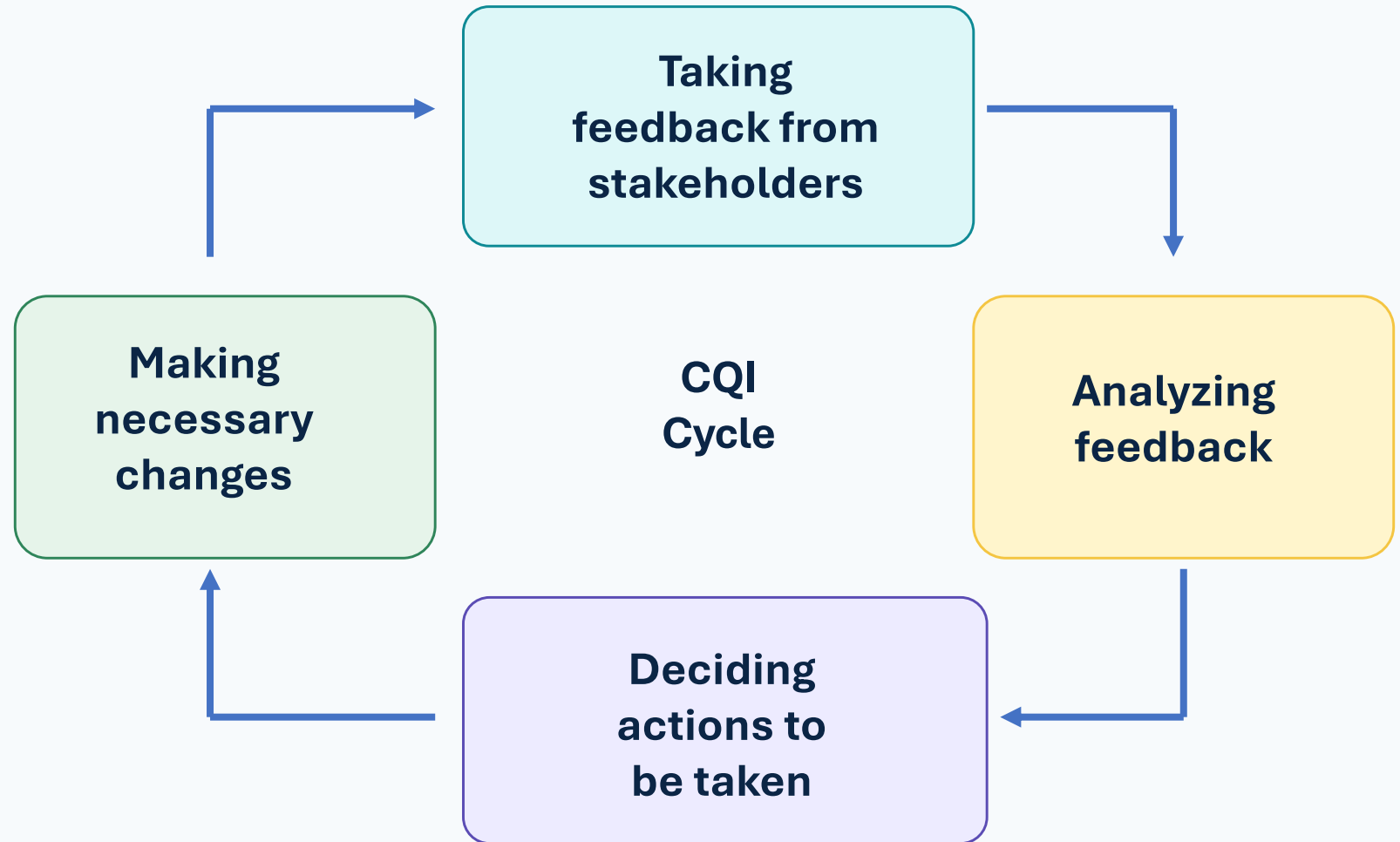
- Practice of appropriate pedagogical methods
- Assessment of CO attainment in core program courses using appropriate assessment tools
- Assessment of PO attainment from CO assessment
- Completion of culminating Capstone / Final Year Design project

CQI Evidence

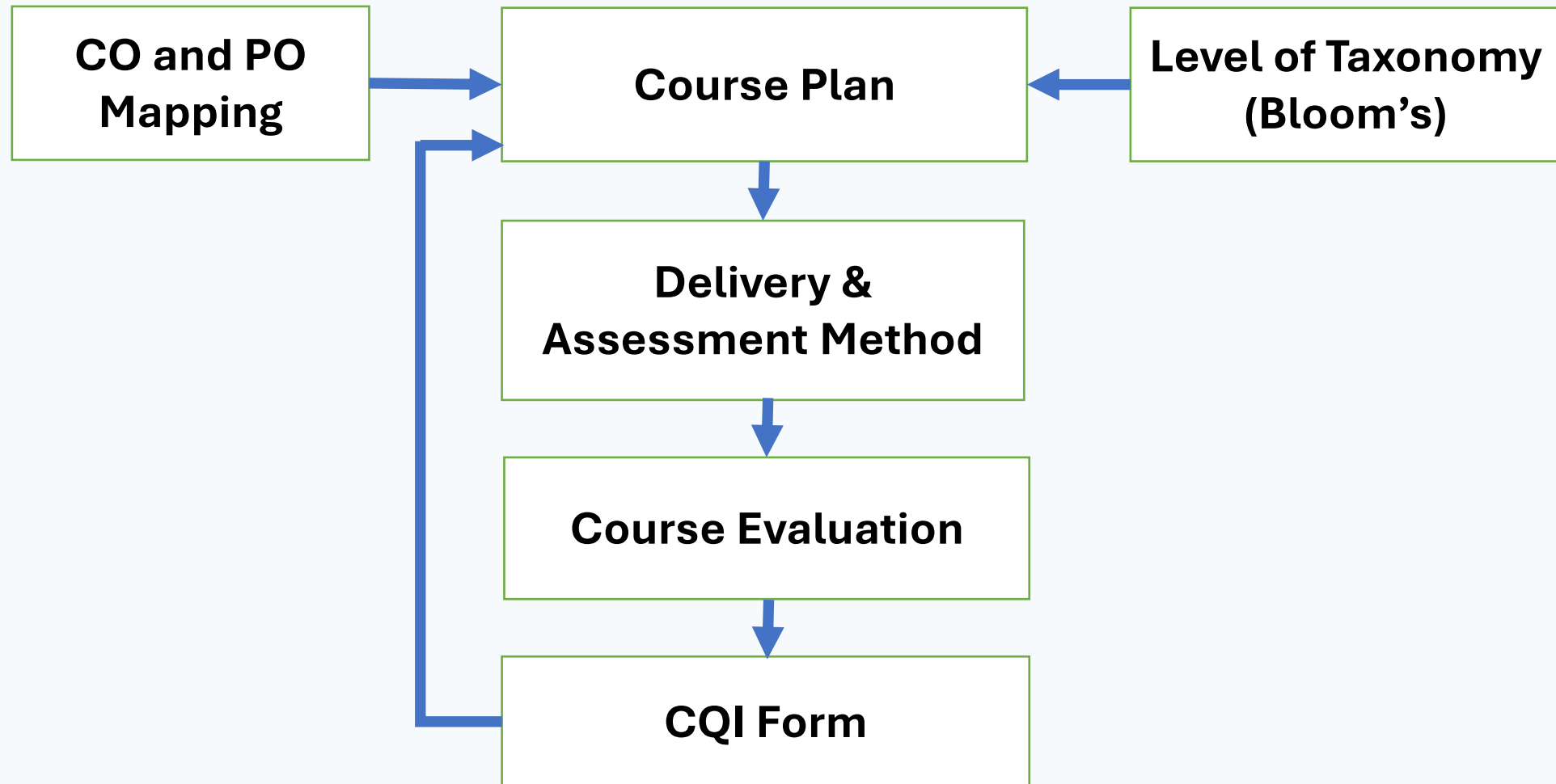
- Use data and feedback to improve processes, policies and facilities

Establish CQI Cycle: Summarize

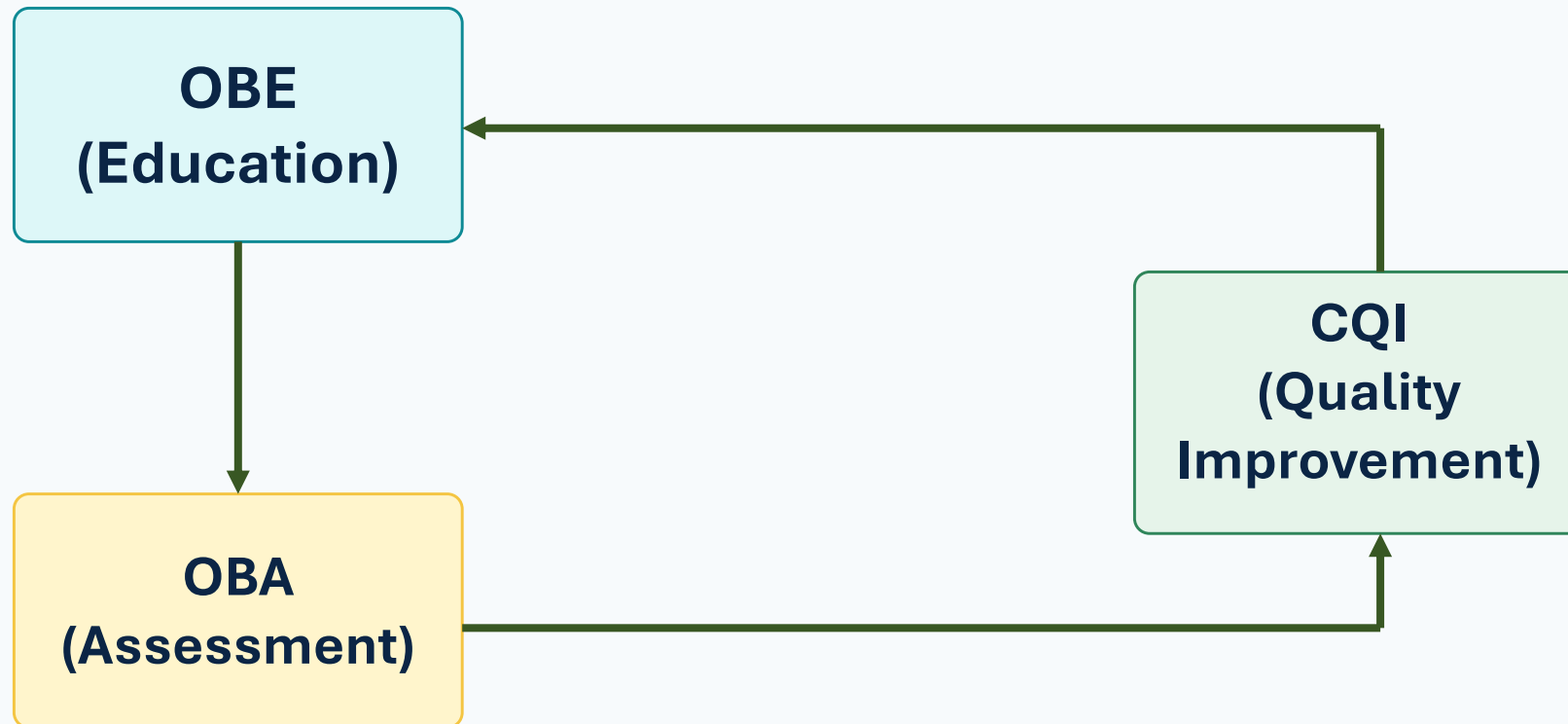
- Fixing a period after which actions will be taken for improvement
- For CO, PO and PEO, different duration is possible



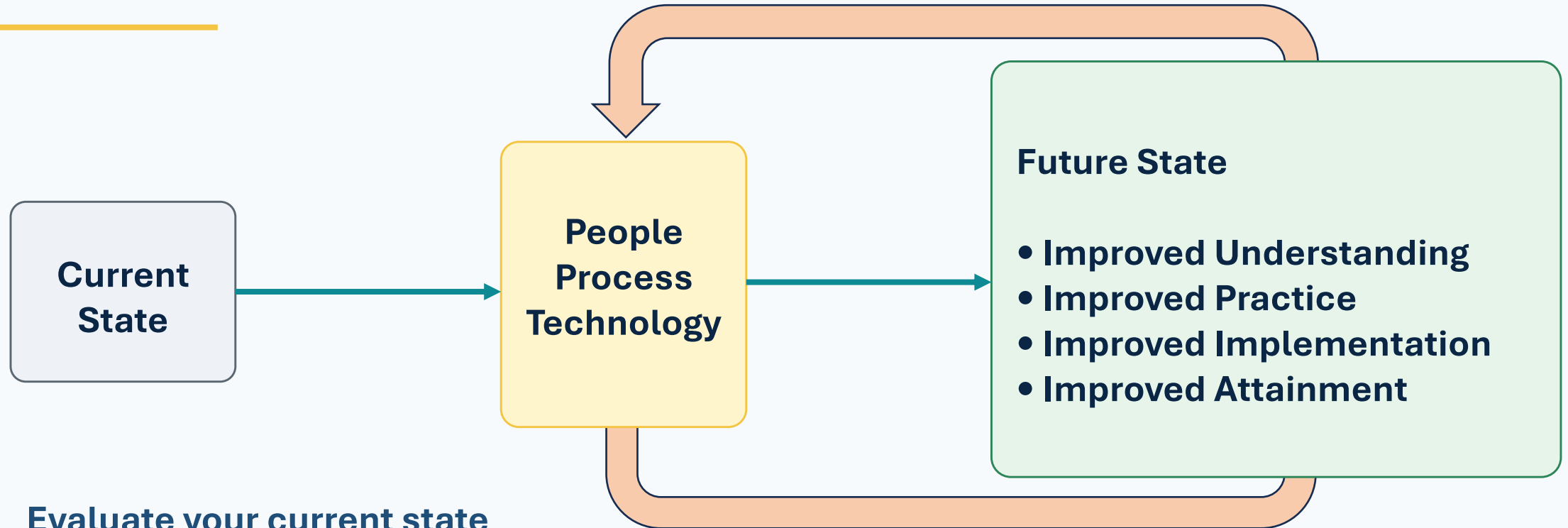
Example: CQI at Course Level



CQI Process: The Link



CQI Exercise



- Evaluate your current state
- Identify the bottlenecks / challenges
- Propose how to fulfill the gaps to close CQI loop

End of Module

Module 5: CQI

Thank You for Your Patience

Question-Answer Session
